

Josh Visser

Toronto, ON | jvisse02@uoguelph.ca | +1 647 542 4431 | linkedin.com/in/josh-visser

Research Interests: Biomedical Applications, Machine Learning, Electromagnetics and Wireless Sensing

Education

Bachelor of Engineering, *Biomedical Major, University of Guelph*

Sept 2020 – May 2025

- GPA: 3.71/4.0 (83.9%)
- GRE: 167 Quant, 161 Verbal, 4 Analytical Writing
- Relevant Coursework: Signal Processing, Systems and Control Theory, Biomedical Signals, Micro & Nano-Scale Electronics, Bio-Instrumentation Design

Research Experience

Machine Learning & Radar Research, *University of Waterloo, Wireless Sensors & Devices lab*

May 2024 – Jan 2025

- Developed a radar-based monitoring system for activity recognition in long-term care homes.
- Improved system classification accuracy to 95% through deep learning model development & optimization.
- Led data collection with 10+ participants and designed preprocessing pipelines for radar data.

AI Medication Assistant & Smart Pillbox, *University of Guelph, Undergraduate Project*

Sept 2024 – April 2025

- Designed and prototyped a computer-vision-based pillbox for medication adherence verification.
- Developed an LLM-based interface to monitor drug interactions, symptoms, and dosage compliance.

Awards, Grants, & Scholarships

- **Rising Stars Research Fellow**, University of Waterloo (2024)
- **MITACS Accelerate Intern** (2024)
- **3rd Place & Semifinalist**, IEEE AP-S Student Design Contest (2025)
- **Dean's Honour List**, University of Guelph — All Semesters (2020–2025)
- **Undergraduate Entrance Scholarship**, University of Guelph (2020)

Publications

- Visser, J.; Abedi, H.; Karmani, S.; Shaker, G. (2025). *AI-Based Activity Monitoring in Washrooms Using Low-Resolution Radar*. IEEE LAMC.
- Hall, J.; Bae, J. B.; Visser, J.; Abedi, H.; Shaker, G. (2024). *Validating mm-Wave Radar System for Fall Detection in Washrooms*. IEEE HI-POCT.
- Visser, J.; Abedi, H.; Shaker, G. (2025). *Activity Monitoring System for Long-Term Care Bathrooms Using mm-Wave FMCW Radar*. IEEE Sens. J. (Submitted).
- Bagheri, M. O.; Chow, J.; Visser, J.; Leong, V.; Shaker, G. *Millimeter-Wave Multi-Radar Tracking System Utilizing a Modified GRIN Luneburg Lens for Real-Time Healthcare Monitoring*. IEEE Trans. Antennas Propag. (Submitted).
- Bagheri, M. O.; Chow, J.; Visser, J.; Leong, V.; Shaker, G. *Synchronized Multi-Radar Fusion System for Real-Time Presence Detection*. IEEE Rad. Conf. (Submitted).
- Mungal, A.; Abedi, H.; Visser, J.; Shaker, G. *Detecting Distracted Drivers Using mm-Wave FMCW Radar and AI* IEEE Sens. J. (In preparation).

Presentations

AI-Based Activity Monitoring in Washrooms Using Low-Resolution Radar,

Jan 2025

IEEE LAMC, San Juan, Puerto Rico

- Presented research findings & system design to audience of 30+ people

Millimeter-Wave Multi-Radar Tracking System Utilizing a Modified GRIN Luneburg Lens for Real-Time Healthcare Monitoring.

July 2025

IEEE AP-S Student Design Contest, IEEE Antennas and Propagation Conference

- Presented technical poster and system design to judges and peers; Competitive student contest.

Skills

Software: Python (Keras, TensorFlow, OpenCV, NumPy, SciKit), MATLAB, C (Basic), SolidWorks, LTspice, Altium

Hardware: Digital Circuit Design, PCB Layout, Rapid Prototyping (soldering, 3D-Printing), Debugging

Work Experience

Hardware Engineer, *Evertz Microsystems, Burlington, ON*

Jan 2025 – May 2025

- Design and testing of high-speed IP router PCBs for professional broadcast equipment.
- Conducted power integrity simulations validated by hardware measurements.
- Resolved 30+ component sourcing and design issues in product power supply circuitry.

R&D Engineer, *Accelovant Technologies, North Vancouver, BC*

Jan 2023 – Aug 2023

- Developed custom SFP modules and temperature probes for fiber-optic temperature sensing system.
- Automated drawing creation with a Visual Basic macro, generating over 50 new SolidWorks documents.
- Designed a high-throughput Bed-of-Nails test fixture improving quality control for 5,000+ PCBs annually.

Leadership & Volunteering

Mechanical Engineer, *University of Guelph Robotics Team*

Sept 2022 – May 2024

- Led redesign of robotic arm for Canadian International Rover Competition.

Engineering Peer Helper, *University of Guelph*

Sept 2021 – May 2022

- Mentored engineering students in organization, time management, and teamwork.